



KONE Elevate 2026

KONE Idea Proposal

— Autonomous Fault Isolation & Root Cause Analysis Assistant

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Table of contents:

Section	Title	Page number:
1	Executive Summary	1
2	The Problem	2
3	Our Proposed Solution	5
4	Architecture & Agent Design	8
5	Fault Detection, Alarm Correlation & RCA	13
6	Future Roadmap	18

Executive Summary

Elevator faults often trigger multiple related alarms, making it difficult for technicians to distinguish the actual root cause from the resulting symptoms, leading to longer downtime, repeat service visits, and unnecessary part replacement.

We propose an Autonomous Fault Isolation & Root Cause Analysis (RCA) Assistant that analyses elevator telemetry, fault logs, and maintenance history to correlate related alarms and identify the most likely root cause. The system evaluates multiple hypotheses using available evidence, provides confidence levels, explains why alternatives were ruled out, and recommends appropriate corrective actions.

The solution provides evidence-backed and auditable diagnostics for technicians, along with a simplified status for building managers. Unlike a simple fault-code monitoring system, it focuses on explaining why a fault occurred, rather than only reporting that a fault occurred.

The AI is strictly advisory and has no autonomous control over the elevator's motor, brake, door locks, safety circuit, or other safety-critical functions. Our solution / proposed idea is currently at the architecture and design stage. Prototype development and physics-grounded synthetic scenario testing are planned as the next steps.

The Problem

Elevator faults often generate multiple interconnected alarms and symptoms, making it difficult for technicians to distinguish the actual root cause from its consequences. For example, if debris blocks an elevator door from closing, the system may register a “door obstruction” fault, followed by a “door close timeout,” a safety-circuit trip, and eventually a “drive start failure” because the motor is safety-locked. These events may appear as separate faults even though they originate from a single underlying issue. This type of alarm cascade can obscure the actual root cause and make diagnosis more difficult.

Identifying the cause currently requires manual correlation of telemetry, fault logs, and maintenance information, which can be time-consuming when several subsystems are involved. This can lead to delayed or incorrect diagnosis, longer downtime, repeat service visits, and unnecessary component replacement.

The key challenge is to move from fault detection to reliable, evidence-based fault isolation and root-cause identification.

Illustrative Real-World Diagnostic Scenario

Consider an elevator that cannot start because its doors fail to close properly. The system may generate several alarms in sequence. For a technician, the challenge is determining which event initiated the fault and which alarms are only consequences.

Illustrative Scenario

Illustrative engineering scenario; not based on a specific KONE service case.



Today vs. With the Proposed Assistant

Today

Manual, time-consuming and expert-dependent.



Multiple alarms



Manual correlation



Review logs & telemetry



Check maintenance history



Technician investigation



Root-cause identification

From
Manual
Investigation
to AI-Assisted
RCA

With the Proposed Assistant

Structured, evidence-based and consistent diagnostic support.



Multiple alarms



Automatic alarm correlation



Fault episode formation



Subsystem isolation



Evidence-based RCA



Confidence-ranked diagnosis



Recommended action



Technician review & action

Real-World Pain Point → AI Capability → Practical Benefit



Real-World Pain Point



Multiple alarms obscure the initiating fault



Fault information is spread across different sources



Several components can produce similar symptoms



Diagnosis can be difficult to verify



Knowing the cause does not automatically tell the technician what to investigate next



Different users need different levels of information



AI Capability



Alarm correlation & fault-episode formation



Multi-source evidence retrieval



Competing hypotheses + fault-tree/Bayesian reasoning



Evidence + confidence + reasoning trace



Corrective-action synthesis



Dual-audience outputs



Practical Benefit



Gives the technician a consolidated view of related events



Brings relevant telemetry, logs, maintenance history and documentation together



Supports structured comparison of possible causes



Makes the diagnostic conclusion easier to review



Provides a practical next step for technician review



Provides detailed diagnostic support for technicians and concise status for building managers

Our Proposed Solution

The **Autonomous Fault Isolation & Root Cause Analysis (RCA) Assistant** works alongside the elevator's existing telemetry, alarm, and maintenance systems. Instead of treating every alarm as a separate fault, the system **correlates related events, identifies the affected subsystem, and determines the most likely underlying root cause.**

The solution will:

- **Correlate alarms and events** occurring within the same time window to identify potential fault cascades.
- **Analyse multiple evidence sources**, including sensor/telemetry data, fault logs, maintenance history, and relevant manuals or fault-code information.
- **Generate and compare multiple root-cause hypotheses** using an elevator-specific fault tree and Bayesian reasoning, rather than relying only on individual fault codes.
- **Rank possible causes with confidence levels** and show the evidence supporting the leading diagnosis, including why alternative causes were ruled out.
- **Recommend corrective actions** based on the identified root cause and available maintenance knowledge.
- Provide **two levels of output**: a detailed, evidence-backed diagnosis for technicians and a simplified status summary for building managers.
- Maintain an **explainable and auditable reasoning trail**, so technicians can review how the system arrived at its conclusion. The system is designed as an **advisory diagnostic layer only**. It will not control or directly operate safety-critical elevator functions; final decisions and physical interventions remain with qualified human personnel.

Key Differentiators

What makes our RCA Assistant different, and the value it brings.

01



Alarm Detection → Causal Fault Isolation

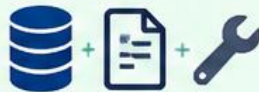
Connects related alarms to identify the underlying cause, not just individual fault codes.



Outcome

Helps technicians focus on the root fault instead of the alarm cascade.

02



Single Signal → Multi-Evidence Diagnosis

Combines telemetry, alarms, maintenance history, technical data, and technician inputs for diagnosis.



Outcome

Enables better-supported and more reliable fault diagnosis.

03



Fixed Rules → Competing Root-Cause Hypotheses

Compares multiple possible causes using elevator-specific fault relationships and reasoning.



Outcome

Improves fault isolation when multiple causes can produce similar symptoms.

04



Diagnosis → Evidence & Confidence

Shows the leading cause, supporting evidence, confidence, and alternative causes considered.



Outcome

Makes the diagnosis transparent, reviewable, and easier for technicians to trust.

05



RCA → Actionable Technician Support

Converts the diagnosis into recommended corrective actions for technician review, with simplified status for building managers.



Outcome

Helps technicians move from “what is wrong?” to “what should be investigated next?”

06



AI Assistance → Explainable & Safety-Bounded

Provides an auditable reasoning trail while keeping safety-critical control and final maintenance decisions with qualified personnel.



Outcome

Enables responsible AI-assisted diagnosis without compromising safety or human oversight.



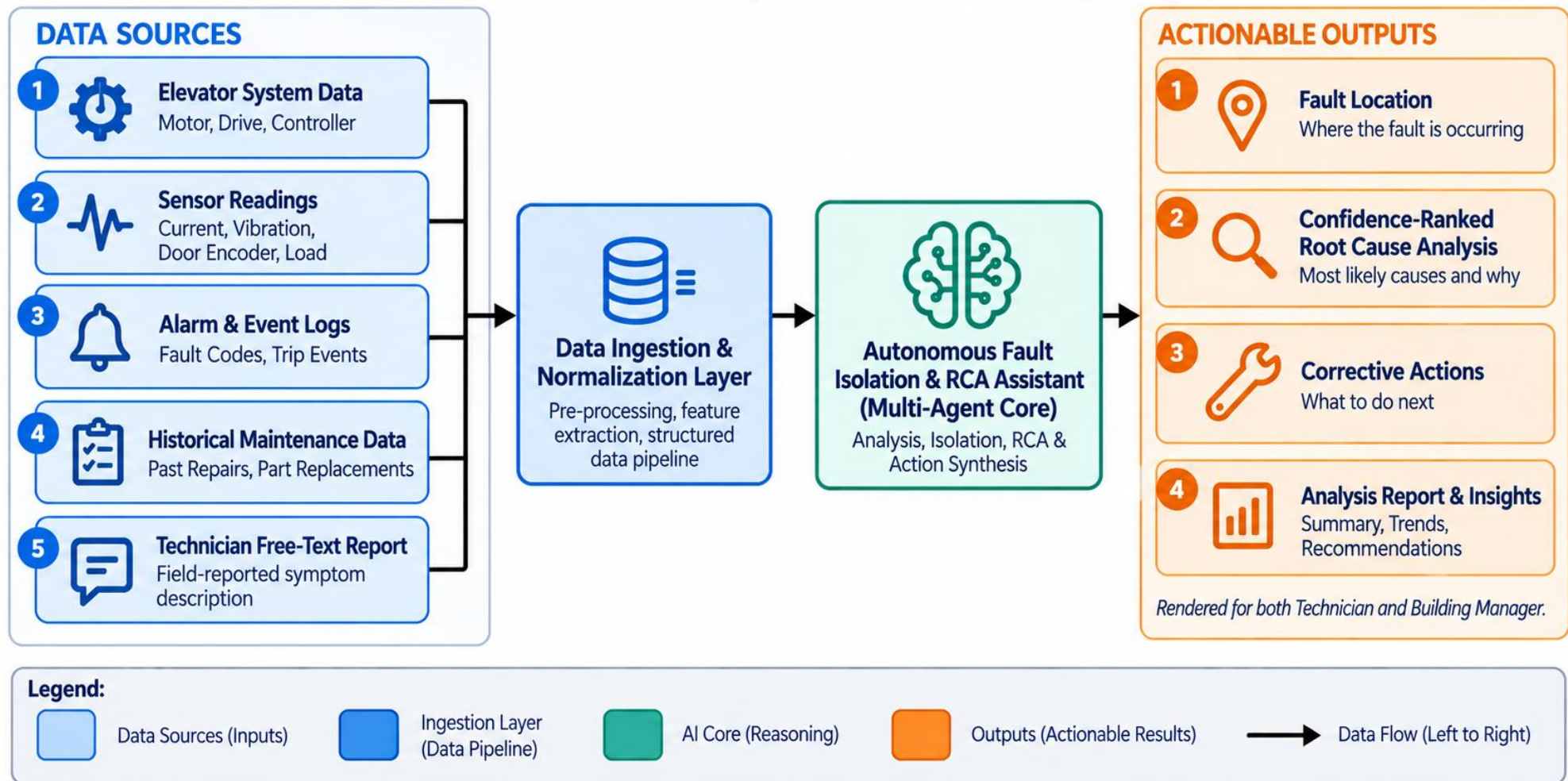
From scattered alarms to evidence-backed insights for faster, more confident maintenance decisions.

System Context: Data Sources to Actionable Output

The system combines **elevator telemetry, sensor readings, alarm/event logs, maintenance history and technician-reported symptoms**. These inputs are **normalized and analyzed** by the **fault-isolation and RCA pipeline** to produce **evidence-backed diagnostic** and actionable outputs for **technicians and building managers**.

System Context: Data Sources to Actionable Output

From real-world elevator data to insights for technicians and building managers



Architecture & Agent Design

System Overview

The system follows a multi-stage, multi-agent architecture that converts elevator data and fault events into an evidence-backed root-cause diagnosis and recommended corrective action.

The architecture begins with multiple elevator data sources, followed by **data ingestion, alarm correlation and triage, orchestration, evidence retrieval, fault isolation** and **root-cause analysis**, and **corrective-action synthesis**. The final result is presented to a human through separate Technician and Building Manager views.

An **Explainability Engine** works across the diagnostic process to capture the **evidence, confidence scores**, and **reasoning trace behind each investigation**.

The entire **AI pipeline is advisory only**. It has **no autonomous control** path to the elevator's safety circuit or other safety-critical hardware.

Data Sources

The system combines both machine-generated and human-generated information.



Elevator System Data

Motor, drive and controller information.



Sensor Readings

Current, vibration, door-cycle timing, encoder/position and load data.



Alarm & Event Logs

Fault codes, warnings and trip events.



Historical Maintenance Data

Previous repairs and component replacements.



Technician Free-Text Reports

Field-reported symptoms and observations.

Agent Design

Component	Main Responsibility
 Alarm Correlation & Triage	Detects abnormal signals and groups related alarms into one fault episode.
 Orchestrator Agent	Coordinates the investigation workflow and passes information between stages.
 Retrieval Agent	Retrieves relevant CMMS records, fault logs and technical manuals.
 RCA Agent	Isolates the affected subsystem and ranks possible root causes using fault-tree/Bayesian reasoning.
 Synthesis Agent	Converts the RCA result into confidence-ranked corrective actions and user-specific reports.
 Explainability Engine	Records evidence, confidence scores and the reasoning trace throughout the investigation.
 Human Review / Sign-off	Allows the technician to review and confirm the recommended action.

Diagnostic Flow



Key Outputs



Fault Location

Identifies the location of the fault within the elevator system.



Confidence-Ranked Root Cause

Provides the most likely root cause(s) with confidence levels.



Supporting Evidence & Reasoning

Shows relevant evidence and the reasoning behind the diagnosis.



Corrective Action

Recommends actionable next steps for resolution.



Technician View

Provides a detailed diagnostic report for technicians.



Building Manager View

Provides a simplified status summary for building managers.

Explainability & Safety

The **Explainability Engine** maintains a trace of the evidence, confidence scores, and reasoning used during each investigation, allowing the technician to understand how the diagnosis was reached.

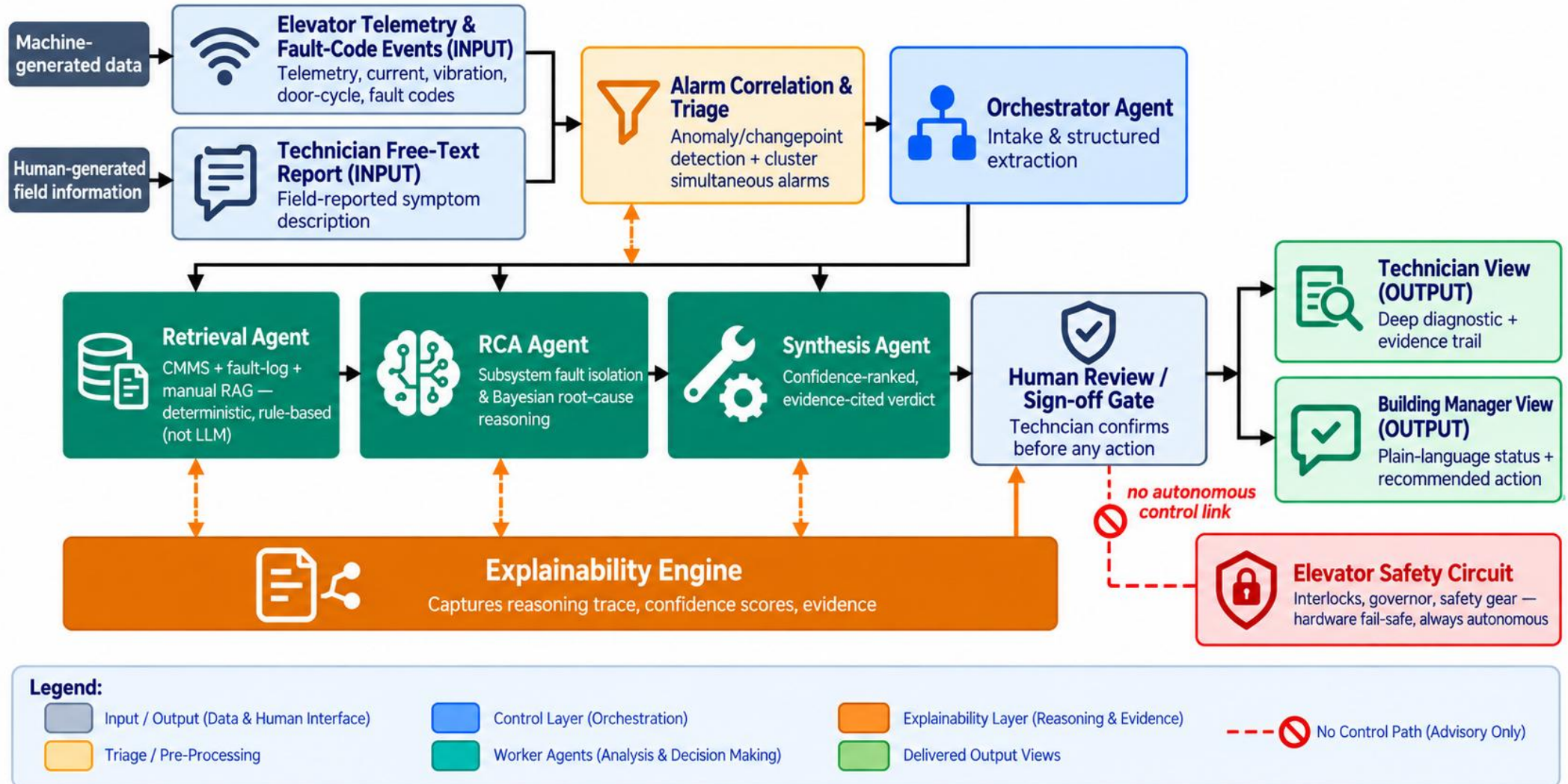
The system is **advisory only**. It has no autonomous control over the elevator's motor, brake, door locks, safety circuit, or other safety-critical functions. The final maintenance decision remains with the qualified technician.

The **solution** combines **deterministic event processing** and **retrieval with AI-assisted reasoning**, rather than relying on a single general-purpose LLM for diagnosis.



Multi-Agent Fault Isolation & RCA Architecture: The following architecture illustrates how the individual components work together to perform the complete fault-isolation and root-cause analysis workflow.

Multi-Agent Fault Isolation & RCA Architecture



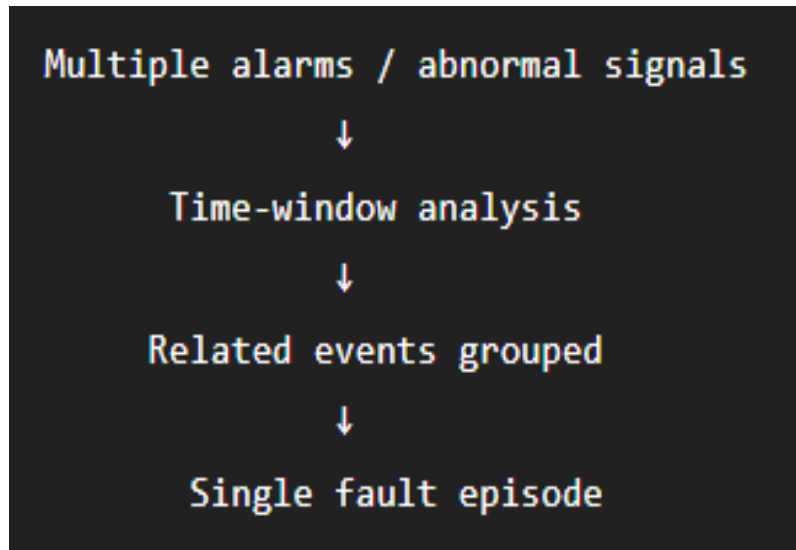
Fault Detection, Alarm Correlation & RCA

Fault Detection

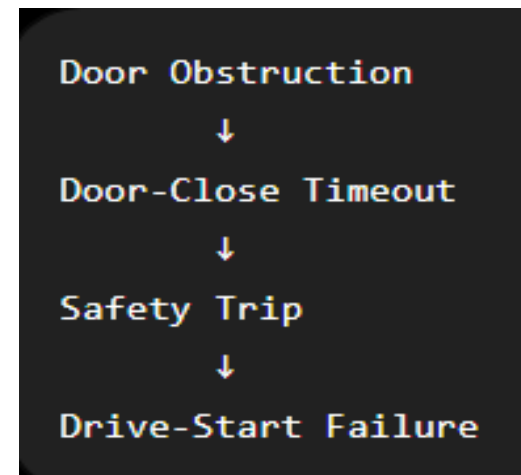
- Monitors **telemetry, sensor readings and fault events**.
- Detects abnormal behaviour using **threshold and change-point analysis**.
- Considers signals such as motor current, vibration, door-cycle timing, encoder/position, load and fault events.
- Triggers an investigation when a significant abnormal condition is detected.

Alarm Correlation

The system groups alarms and abnormal signals that occur close together in time into a **single fault episode**, rather than treating every alarm as an independent fault.



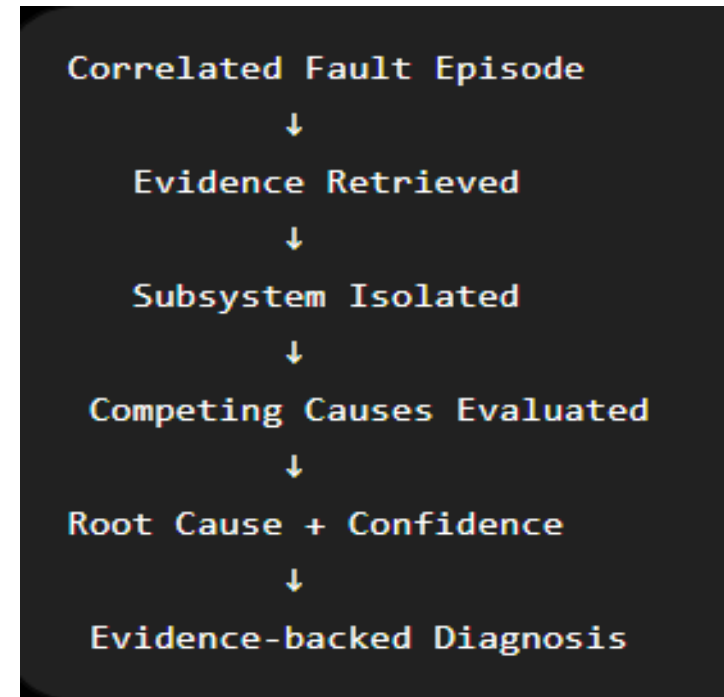
Example:



Root Cause Analysis (RCA)

Once the fault episode is formed, the **RCA Agent**:

- **Retrieves** relevant maintenance **records**, **fault logs** and technical documentation.
- **Identifies** the **affected elevator subsystem**.
- Generates **possible root-cause hypotheses**.
- Uses **fault-tree** and **Bayesian reasoning** to compare the possible causes.
- Ranks the causes using **confidence** and **supporting evidence**.
- Identifies **alternative** causes and **explains** why they are less likely.



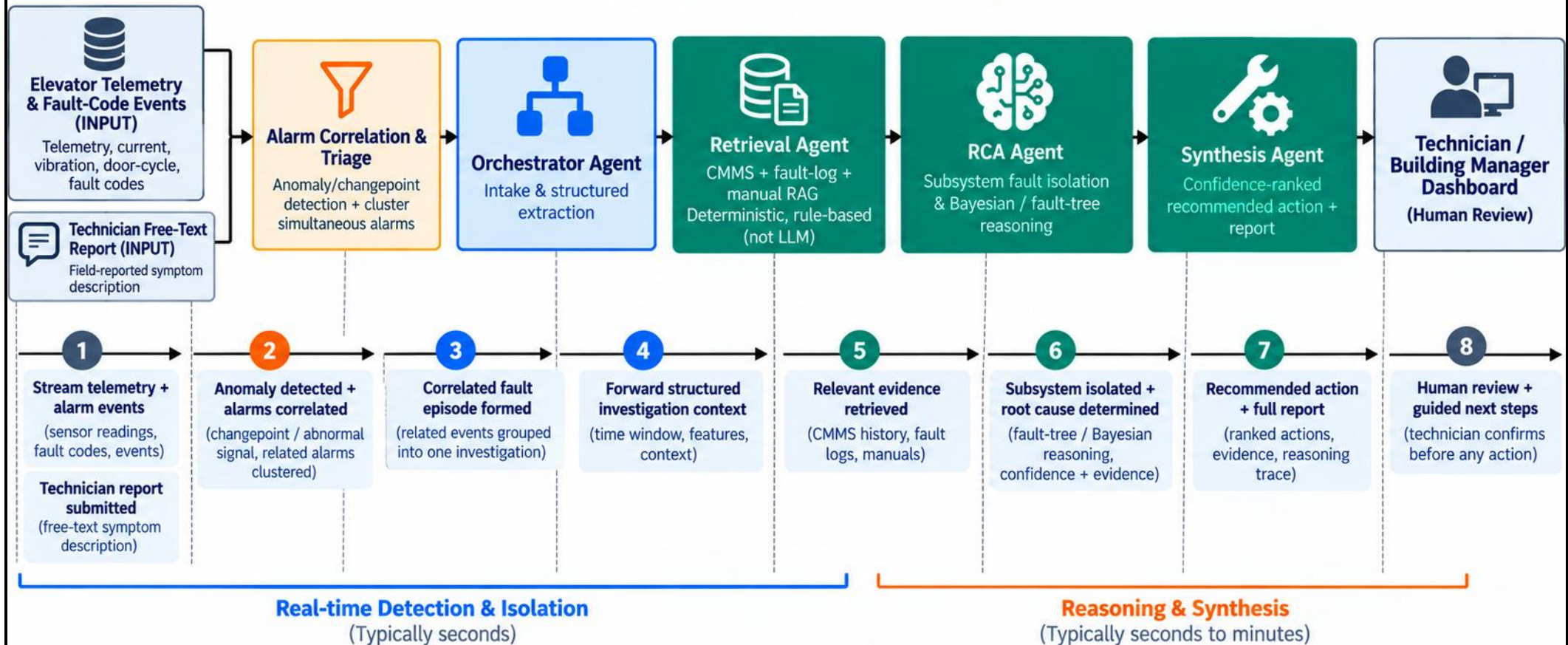
RCA Objective:

Move from “*multiple alarms occurred*” to “*these events are connected, this subsystem is affected, and this is the most likely underlying cause*” — supported by evidence and confidence.

End-to-End RCA Sequence Flow: The following sequence flow illustrates how a fault event moves through each stage of the system, from initial detection and alarm correlation to root-cause analysis, recommended action, and final human review.

End-to-End RCA Sequence Flow

From elevator fault event to actionable output



Legend:

Input / Output
(Data & Human Interface)

Control Layer
(Orchestration)

Worker Agents
(Analysis & Decision Making)

Response / Feedback
(Within same component)

Data Flow
(Left to Right)

Notes:

- All connections are internal software data/reasoning flow only.
- No control authority over elevator hardware.
- Timing is illustrative and may vary by deployment.

Current Assumptions & Limitations

The proposed solution is currently at the **architecture** and **design stage**. The following assumptions and limitations will be addressed during prototype development and KONE SME review:

- **KONE-specific knowledge:** Fault relationships, fault-code mappings, and diagnostic rules shown are illustrative and will be refined using KONE domain expertise and documentation when available.
- **Data availability:** The proposed approach assumes access to relevant elevator telemetry, alarm/event logs, maintenance history, and technician observations. Actual data availability and quality will need to be assessed.
- **Initial validation:** Prototype evaluation will use controlled, physics-grounded synthetic scenarios rather than field-validated data.
- **Data uncertainty:** When evidence is incomplete, inconsistent, or conflicting, the system will reduce diagnostic confidence and flag the case for further technician investigation rather than force a conclusion.
- **Safety boundary:** The system is strictly advisory and will not autonomously control or operate safety-critical elevator functions. Final maintenance decisions remain with qualified personnel.

Success Criteria & Evaluation Metrics

The prototype will be evaluated using controlled, physics-grounded synthetic fault scenarios. Evaluation will assess whether the system can reliably connect events, isolate the affected subsystem, identify the likely root cause, and provide evidence-backed and actionable diagnostic support.

Evaluation Area		Success Criterion
 Alarm Correlation		Correctly groups related alarms into a single fault episode
 Fault Isolation		Correctly identifies the subsystem associated with the fault
 RCA Accuracy		Ranks the known root cause among the leading hypotheses
 Evidence & Explainability		Provides relevant, traceable evidence supporting the diagnosis
 Confidence & Uncertainty		Appropriately reflects the strength or limitations of available evidence
 Corrective-Action Relevance		Produces technically relevant next-step recommendations for technician review
 Technician Usefulness		Provides a clear and understandable diagnostic output for human review

Future Roadmap

The solution is currently at the architecture and design stage. The next phase will focus on incorporating KONE domain expertise, developing the proposed prototype, and evaluating the RCA workflow using controlled synthetic scenarios. KONE's SME review and guidance will be an important step in refining the elevator-specific fault taxonomy, alarm relationships, data requirements, RCA logic, and recommended actions.

Key Future Enhancements:

1. **KONE-specific knowledge:** Replace the illustrative fault taxonomy and fault-code assumptions with KONE-provided domain knowledge and documentation when available.
2. **RCA refinement:** Improve alarm relationships, fault-tree/Bayesian reasoning, and confidence estimation using SME feedback and test results.
3. **Data-driven improvement:** Progress from synthetic scenarios toward historical/real-world telemetry and maintenance data when appropriate data becomes available.
4. **Service workflow integration:** Explore integration with maintenance and technician-support workflows as the solution matures.

ROADMAP FLOW

